

Appendix-A.

FORMAT OF THE PROJECT PROPOSAL/SYNOPSIS

1. Title of the Project:

The title of the project is to be meaningful and must convey the tools/ platform used, e.g.,
“Speech Recognition System in ASP.NET”

While selecting the language/platform for development, students are advised to select one where they have a strength.

2. Problem Definition:

Students are required to give a brief description of the system that is proposed to be computerized (e.g., in case of a Speech Recognition System, your understanding of this system must be explained in brief). In addition, students are required to give the exact outcome of the system. i.e., what/which part of the system is proposed to be computerized in the schedule and available expertise?

3. Objectives & Scope: Explain the objectives and the scope of the project.

4. Methodology:

- (a) Explain the methodology for data collection.
- (b) Explain the technique(s) & tools/platform proposed to be used for systems analysis, design, testing, and development of software.

5. Nature of Research: If you are writing the Dissertation based on a literature survey, give details of the literature to be surveyed. Explain whether the nature of the research is exploratory or original thought process is being pursued.

**NOTE:ONCE THE TITLE,LANGUAGE/PLATFORMAREFIXED
THEY CANNOT BE CHANGED.**

METHODOLOGY FOR PROJECT WORK/DISSERTATION

Chapter-1:Introduction/ProblemDefinition

1. The following aspects need to be covered in the first chapter:
 - (a) **Brief Description of the Organization:** If the project work is being done outside the Institute, describe the organization, its nature of business, products, turnover, market position, etc.
 - (b) **General Description of the System under Study:** Briefly explain how the existing information system achieves the task that is under study.
 - (c) **User Requirements:** Explain in non-technical terms why the existing information system is not able to meet the current tasks against the objectives/standards/benchmarks of the organization. Problems faced by the existing system.
 - (d) **Establish the Need for the New System:**
 - (i) Give justification, how, & Why the new system user requirements cannot be fulfilled.
 - (ii) How will the proposed solution enhance the efficiency, effectiveness, control, security, and achievement of objectives/standards/benchmarks?
 - (e) **Objectives of the Project:** Keeping in view the user requirements and needs, available time and expertise, lay down the objectives of the project. Clearly spell out what is proposed to be done in the project.
 - (f) **Methodology:** SDLC, Structured Analysis & Design, Object Oriented Design method, or Prototype method. You may follow a mixed methodology and tools; however, you should clearly mention the steps and tools, and techniques that you are going to follow in the current project with a brief justification.
 - (g) **Data required & Data Collection Method:** You should establish the requirement of data and the methodology/technique of data collection, e.g., interview, questionnaire, document analysis. If a questionnaire is developed, attach a copy of the questionnaire as an appendix.

Chapter 2: Systems Requirement Analysis

2. In this chapter, students are required to establish the user requirements by doing the systems analysis of the existing system in vogue and by interacting with the management/staff of the organization. In every organization, a system does exist in some rudimentary or manual form, if not computerized. The activities that are required to be carried out and to be described in the project are:

- (a) Identify the process (es).
 - (b) Identify the Input to and Output of each process, e.g., for the Rail Reservation System input is the form and the output is a ticket.
 - (c) Identify the data elements (fields) in each input and output.
 - (d) Identify the procedures/rules/mathematical relationships used for converting input into output.
 - (e) Identify the controls (for I/O & access), security needs, validation rules, and codes used for data elements.
 - (f) Identify deficiencies in the existing system.
9. Having collected the data on the above aspects, analyse it and interact with the management for any process changes (es), input, output, procedures, access and control rights, security needs, and your suggestions for removing the deficiencies.

Chapter-3: Systems Design

4. **Physical Design:** Explain relationship between various components (processes, input, output & entities) of the system. Draw DFDs and other diagrams.

5. **Explain Processing Logic:** Using Programmed Flow Charts, Algorithm, Decision Table/Tree, or Pseudo codes.

6. **Interface Design:**

(a) **Output Design:** Screen & Print layouts, i.e., the format and contents (fields) to be included – along with the file to which it is linked.

(b) **Input (Form or Screen) Design:** For keying in data. Give fields, format, codes to be used, validation (error detection) requirements & the file to which it is linked.

7. **Database & File Design:**

(a) **Database Files/Tables Design:** Depending on the DBMS/RDBMS package used, give contents of each file, including Field Name, Type, Size & Description.

(b) **Programme Files or Query Design:** Give the purpose and detailed working of each programme file or a Query and link it to the processing logic and Process/Input/Output.

(c) **Compose Data Element Dictionary:** List all the fields in all the files in alphabetical order in a tabular form and serially number them.

Data Element S No	FIELD NAME	TYPE & SIZE	FIELD DESCRI PTION	FILE NAME/No	PROCESS NAME/No
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Chapter 4: Systems Development

8. **Purpose:** To carry out the activities of writing actual programs, their debugging, testing, and validation. The following activities are to be carried out:

(a) **Program Development:** Explain the language or package used for developing the programmes (both back-end and front-end).

(b) **Testing & Debugging:** Use *Past Data* to check whether the programmes work as intended by

(i) **Module Testing:** Individual program testing.

(ii) **System Testing:** Integrated module Testing for the entire system.

(You may like to include the test reports in the project to show the errors, if any, and a write-up on their rectification.)

(c) **Validation:** Check the system with real-time data for input, output, computational processing & controls.

9. Attach program codes, input and output designs, and outputs using the real data in this chapter. Codes may be attached as an appendix.

Chapter 5: Systems Implementation

10. The following activities are required to be conducted:

- (a) **Acquisition:** Work out the memory requirement, disk space required for programme (s) and database (s). List out HW, SW, and people resources required and indicate the cost of the system.
- (b) **Conversion:** Suggest change over procedures (Parallel, Direct or Partial Change over) with justification.
- (c) **Training Needs:** Establish the training needs of operating personnel, clerical/non-clerical staff, supervisors, and senior staff.
- (d) **Documentation:**
 - (i) Include Operation Manual (instructions for running programmes).
 - (ii) User Manual - Giving instructions & procedures for end-users (for data entry, output, help, etc).